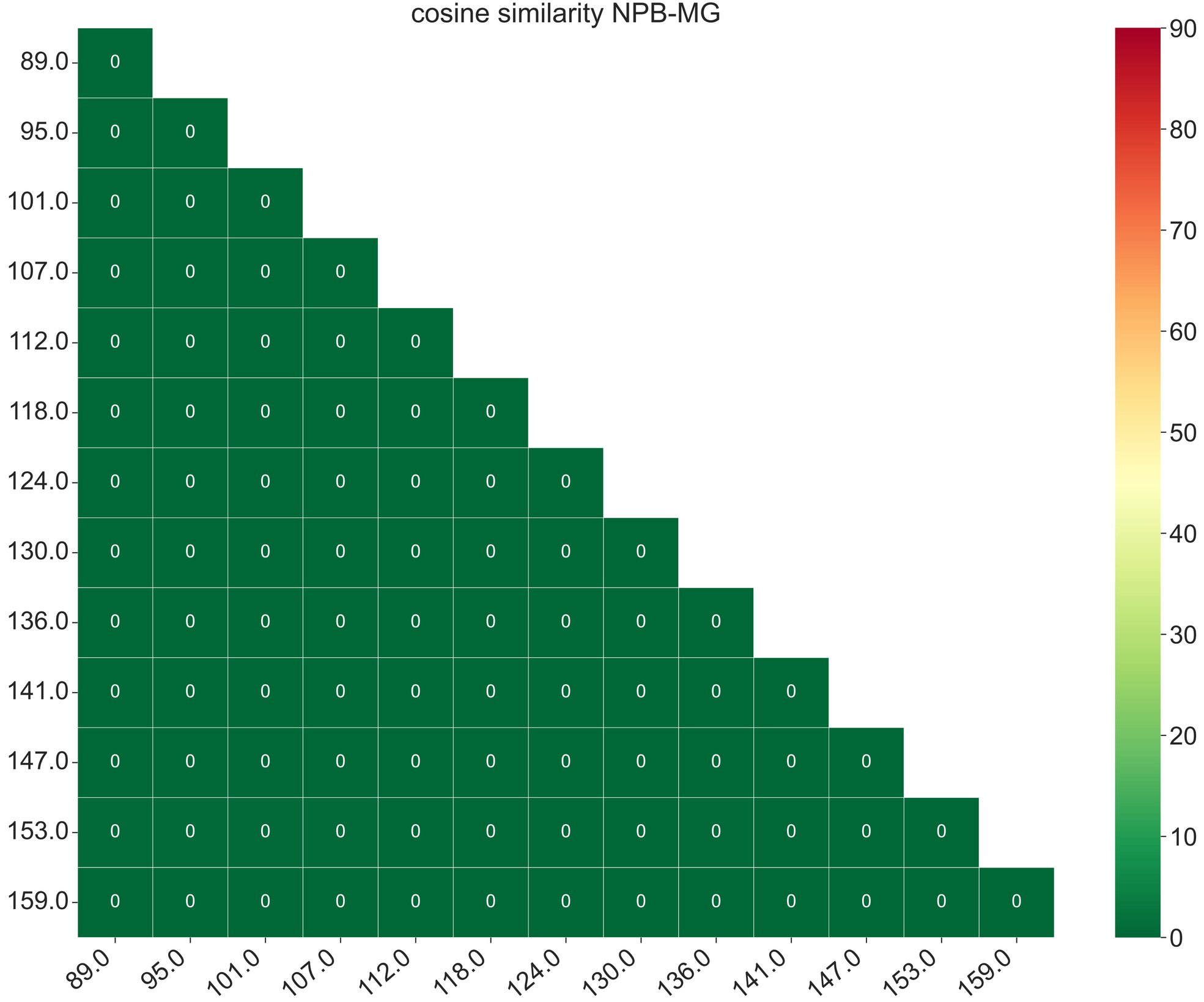cosine similarity NPB-MG